

For the defining equations,

1. You have the variables **PS**, which is for the y-axis, and **DN or V** which is for the x-axis.
2. The equations of the red lines in the graph as are follows
 - a. I will give the **x** and **y coordinates** for the starting point for the line and the end point for the line
 - b. The equation of the line will be given in the form of **PS** for **x coordinate** and **DN or V** for **y coordinate**
3. For the axes
 - a. The axes values are given for the **x and y axes**
 - b. the **x** and **y coordinates** given for the **axes** indicate the equations for a line for the gridlines
 - c. additionally minor gridlines will be given

Graph for vessels-Dangerous gas

Figure 1

Defining Equations (red lines)

PS = 100 000	
X	Y
0,1	100 000
1	100 000

PS*V = 100 000	
X	Y
1	100 000
2 000	50

PS = 20 000	
X	Y
0,1	20 000
1	20 000

PS*V = 20 000	
X	Y
1	20 000
400	50

V = 1	
X	Y
1	2 000
1	20 000

PS*V = 5 000	
X	Y
1	5 000
100	50

PS = 50	
X	Y
0,1	50
100 000	50

PS*V = 2 000	
X	Y
1	2 000
40	50

Axis Values

Y- Axis	
X	Y
0,1	0,1
0,1	1
0,1	10
0,1	100
0,1	1 000
0,1	10 000
0,1	100 000
0,1	400 000

X- Axis	
X	Y
0,1	0,1
1	0,1
10	0,1
100	0,1
1 000	0,1
10 000	0,1

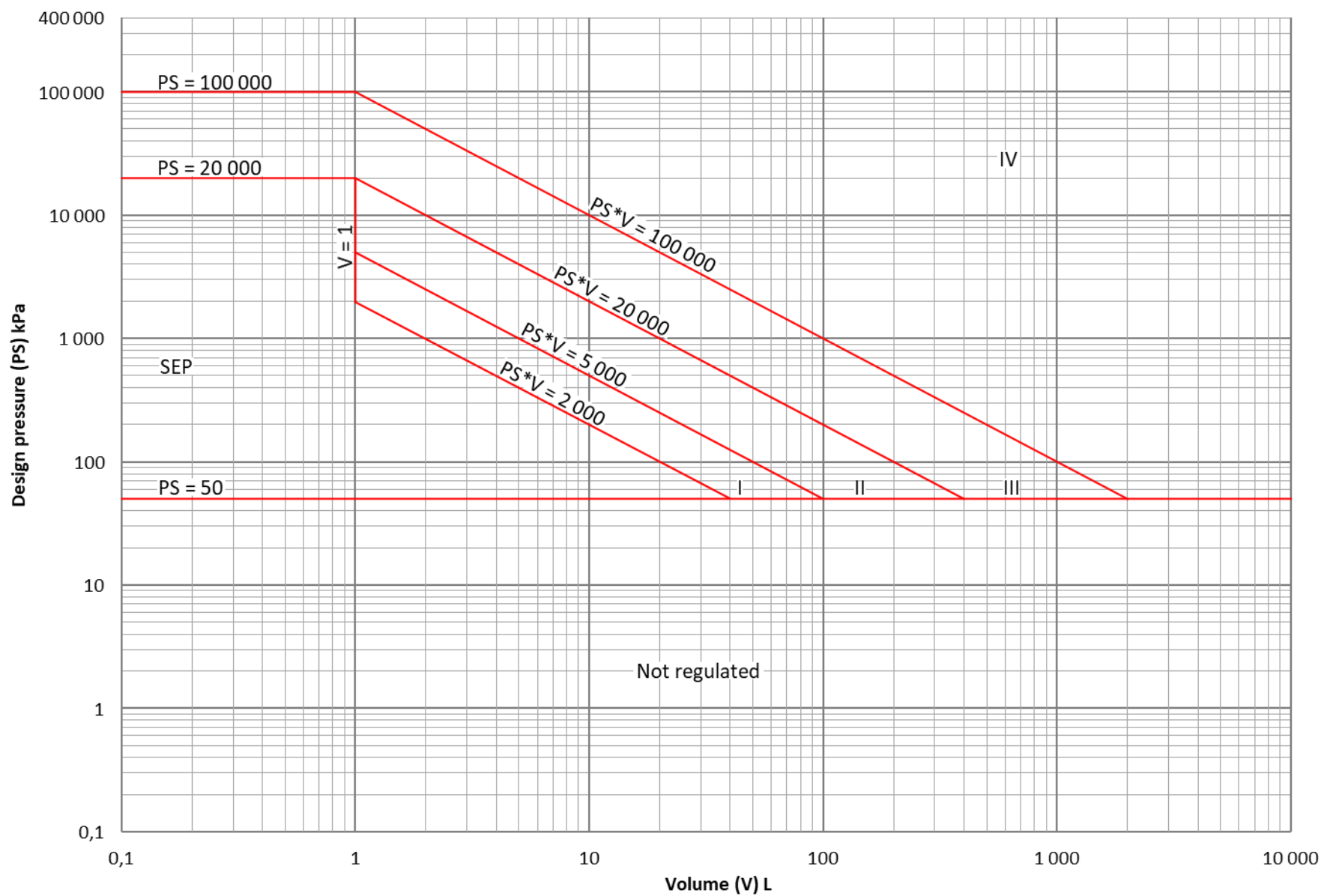


Figure 1 - Graph for vessels - Dangerous gas

Graph for vessels - Non-dangerous gas

Figure 2

Defining Equations

PS = 300 000		PS*V = 300 000	
X	Y	X	Y
0,1	300 000	1	300 000
1	300 000	750	400

PS = 100 000		PS*V = 100 000	
X	Y	X	Y
0,1	100 000	1	100 000
1	100 000	2 000	50

PS = 50		PS*V = 20 000	
X	Y	X	Y
0,1	50	1	20 000
10 000	50	400	50

PS = 400		PS*V = 5 000	
X	Y	X	Y
750	400	1	5 000
10 000	400	100	50

V = 1	
X	Y
1	5 000
1	100 000

Axis Values

Y-Axis		X-Axis	
X	Y	X	Y
0,1	0,1	0,1	0,1
0,1	1	1	0,1
0,1	10	10	0,1
0,1	100	100	0,1
0,1	1 000	1 000	0,1
0,1	10 000	10 000	0,1
0,1	100 000		
0,1	500 000		

Additional Minor gridlines

V = 750	
X	Y
50	0,1
750	500 000

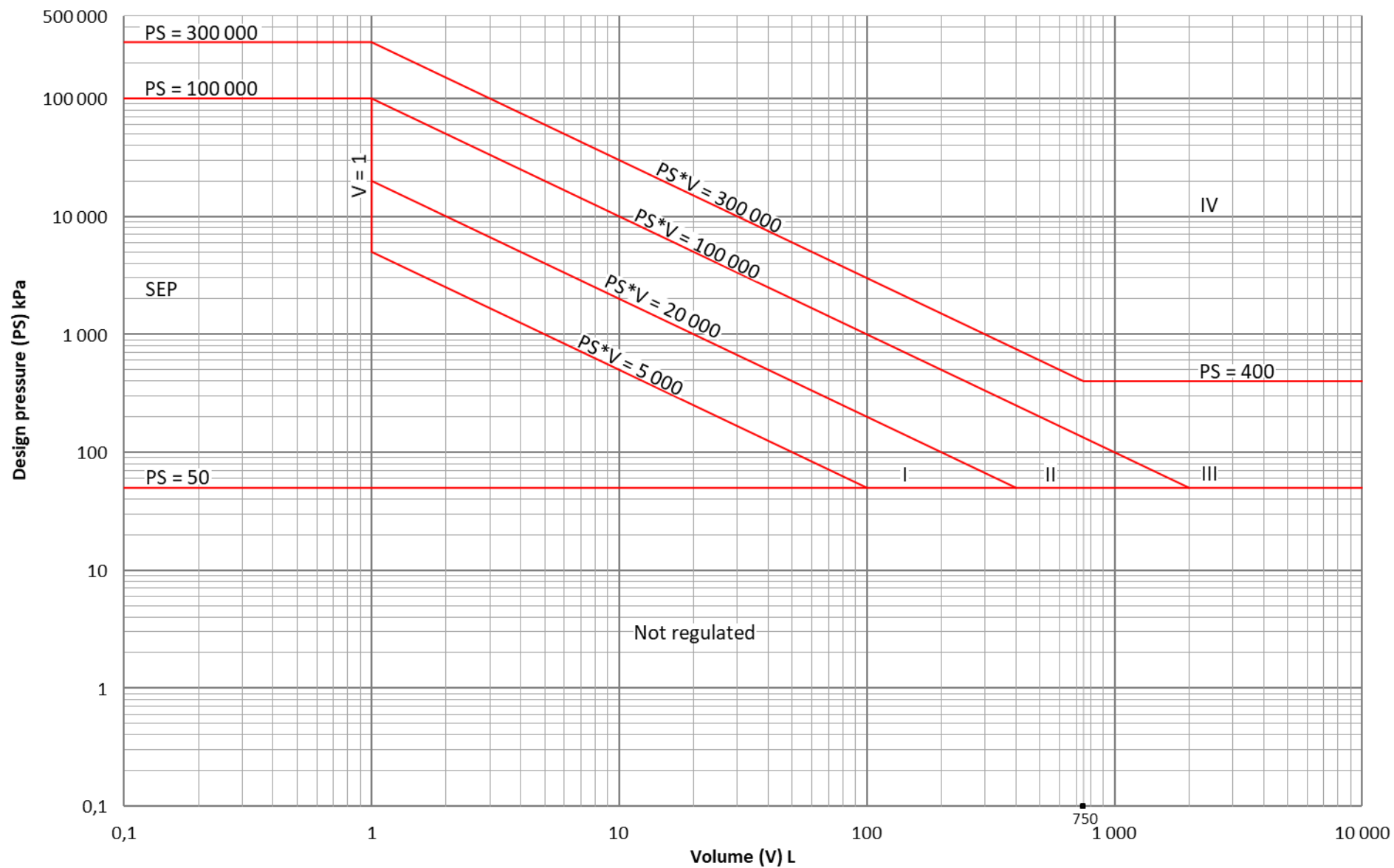


Figure 2 - Graph for vessels - Non-dangerous gas

Graph for vessels -Dangerous liquids

Figure 3

Defining Equations

PS = 50 000	
X	Y
0,1	50 000
10 000	50 000

PS*V = 20 000	
X	Y
1	20 000
400	50

PS = 1 000	
X	Y
20	1 000
10 000	1 000

PS = 50	
X	Y
0,1	50
10 000	50

V = 1	
X	Y
1	500 000
1	20 000

Axis Values

Y-Axis	
X	Y
0,1	0,1
0,1	1
0,1	10
0,1	100
0,1	1 000
0,1	10 000
0,1	100 000
0,1	500 000

X-Axis	
X	Y
0,1	0,1
1	0,1
10	0,1
100	0,1
1 000	0,1
10 000	0,1

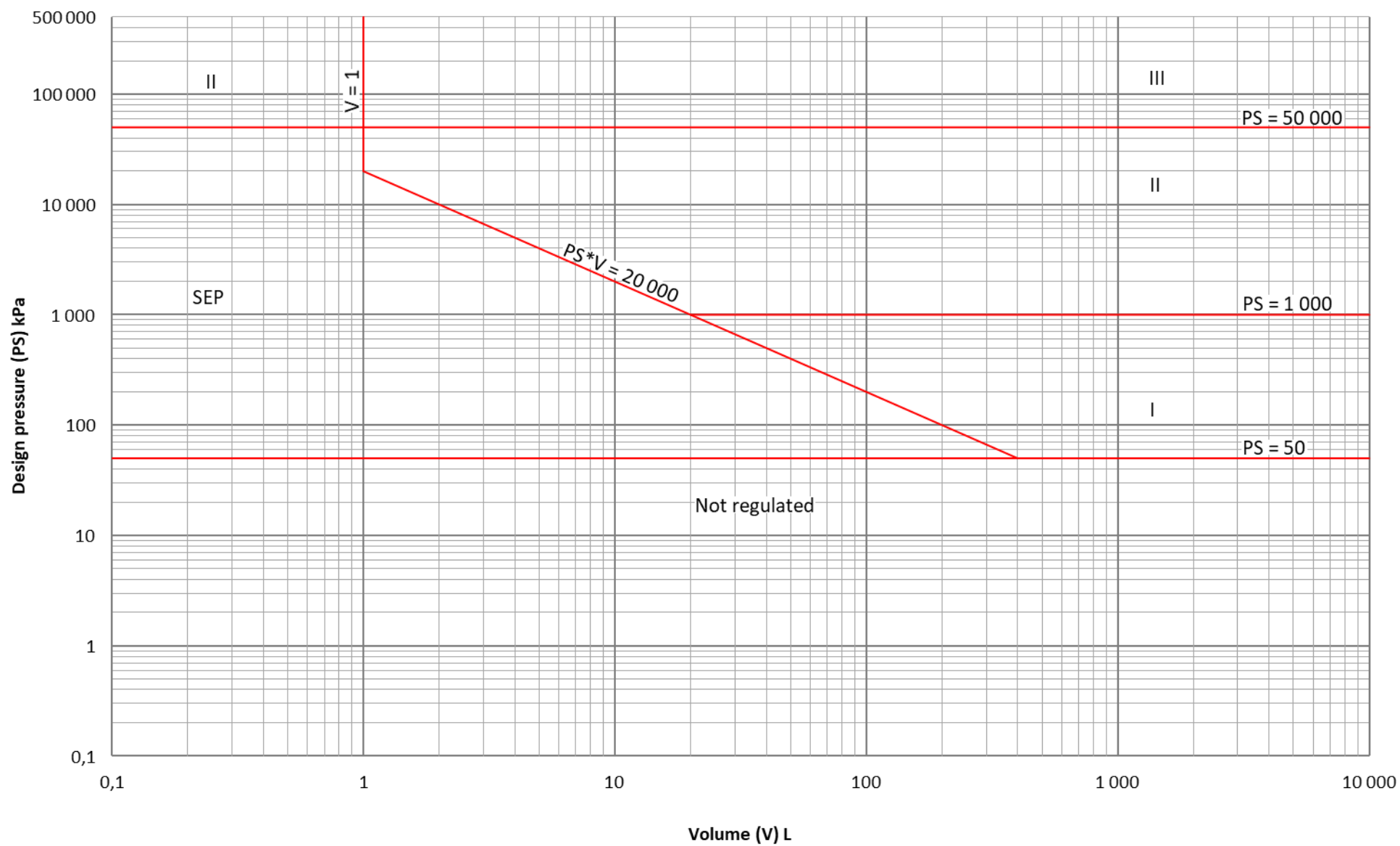


Figure 3 - Graph for vessels - Dangerous liquids

Graph for vessels-Non-dangerous liquids

Figure 4

Defining Equations

PS = 50	
X	Y
0,1	50
100 000	50

PS*V = 100 0000	
X	Y
10	100 000
1 000	1 000

PS = 100 000	
X	Y
0,1	100 000
10	100 000

PS = 1 000	
X	Y
1 000	1 000
100 000	1 000

PS = 50 000	
X	Y
20	50 000
100 000	50 000

V = 10	
X	Y
10	100 000
10	500 000

Axis Values

Y-Axis	
X	Y
0,1	0,1
0,1	1
0,1	10
0,1	100
0,1	1 000
0,1	10 000
0,1	100 000
0,1	500 000

X-Axis	
X	Y
0,1	0,1
1	0,1
10	0,1
100	0,1
1 000	0,1
10 000	0,1
100 000	0,1

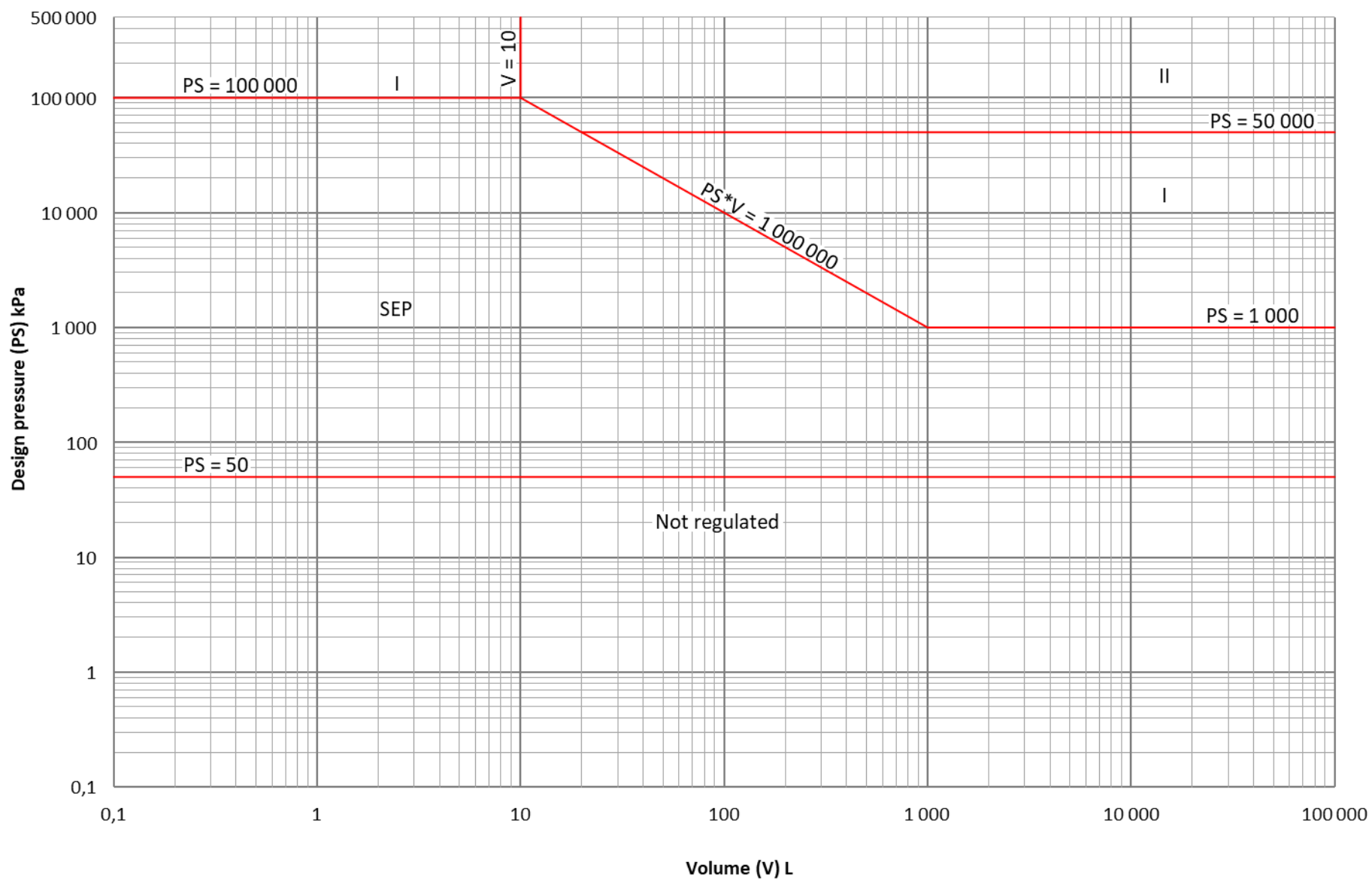


Figure 4 - Graph for vessels - Non-dangerous liquids

Graph for steam generators

Figure 5

Defining Equations

PS = 50		PS*V = 300 000	
X	Y	X	Y
0,1	50	93,75	3 200
10 000	50	1 000	300

PS = 3 200		PS*V = 20 000	
X	Y	X	Y
2	3 200	6,25	3 200
93,75	3 200	400	50

V = 2		PS*V = 5 000	
X	Y	X	Y
2	2 500	2	2 500
2	100 000	100	50

V = 1 000	
X	Y
1 000	50
1 000	300

Axis Values

Y-Axis		X-Axis	
X	Y	X	Y
0,1	0,1	0,1	0,1
0,1	1	1	0,1
0,1	10	10	0,1
0,1	100	100	0,1
0,1	1 000	1 000	0,1
0,1	10 000	10 000	0,1
0,1	100 000		

Additional Minor gridlines

PS = 3 200(GREY)		PS = 2500	
X	Y	X	Y
0,1	3 200	0,1	2500
10 000	3 200	10000	2500

V = 93.75		V= 6.25	
X	Y	X	Y
93,75	0,1	6,25	0,1
93,75	100 000	6,25	100 000

Graph for piping-Dangerous gas

Figure 6

Defining Equations

PS = 50		PS*DN = 350 000	
X	Y	X	Y
1	50	350	1 000
2 000	50	100	3 500

DN = 25(RED)		PS*DN = 100 000	
X	Y	X	Y
25	50	100	1 000
25	100 000	25	4 000

DN = 100(TOP)		DN = 350	
X	Y	X	Y
100	3 500	350	50
100	100 000	350	1 000

DN = 100(BOTTOM)	
X	Y
100	50
100	1 000

Axis Values

Y-Axis		X-Axis	
X	Y	X	Y
1	1	1	1
1	1	10	1
1	10	25	1
1	100	100	1
1	1 000	350	1
1	3 500	1 000	1
1	10 000	2 000	1
1	100 000		

Additional Minor gridlines

PS = 3 500		DN = 350(GREY)	
X	Y	X	Y
1	3 500	350	1
2 000	3 500	350	50

DN = 25(GREY)	
X	Y
25	1
25	50

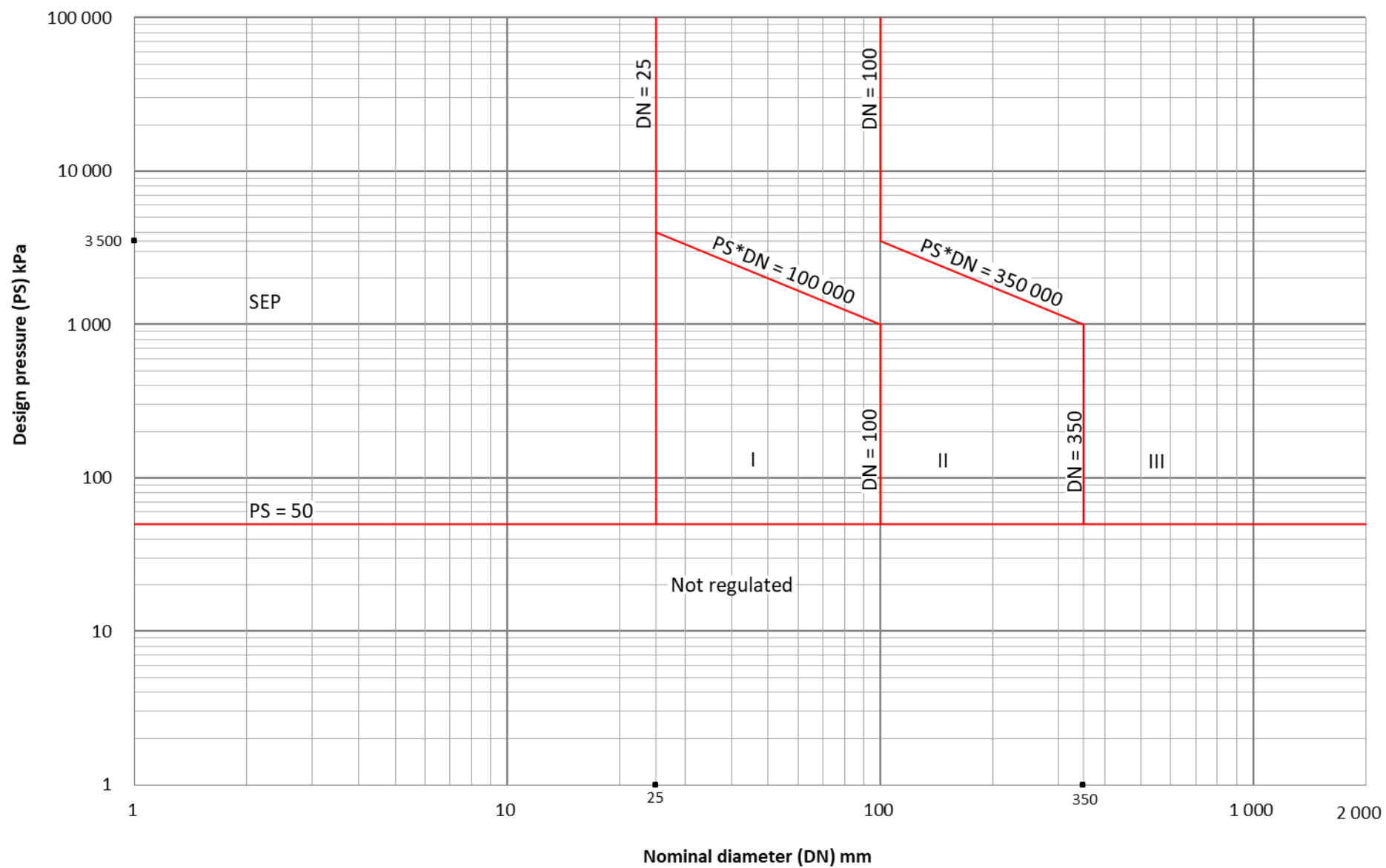


Figure 6 - Graph for piping - Dangerous gas

Graph for piping - Non-dangerous gas

Figure 7

Defining Equations

PS = 50		PS*DN = 500 000	
X	Y	X	Y
0,1	50	250	2 000
20 000	50	10 000	50

DN = 32(RED)		PS*DN = 350 000	
X	Y	X	Y
32	3 125	100	3 500
32	200 000	7 000	50

DN = 100		PS*DN = 100 000	
X	Y	X	Y
100	200 000	32	3 125
100	3 500	2 000	50

DN = 250(RED)	
X	Y
250	2 000
250	200 000

Axis Values

Y- Axis		X-Axis	
X	Y	X	Y
1	1	1	1
1	10	10	1
1	100	100	1
1	1 000	1 000	1
1	10 000	10 000	1
1	100 000	20 000	1
1	200 000		

Additional Minor gridlines

DN = 32(GREY)		PS = 3125	
X	Y	X	Y
32	1	1	3 125
32	200 000	20 000	3 125

DN = 250(GREY)		PS = 3500	
X	Y	X	Y
250	1	1	3 500
250	200 000	20 000	3 500

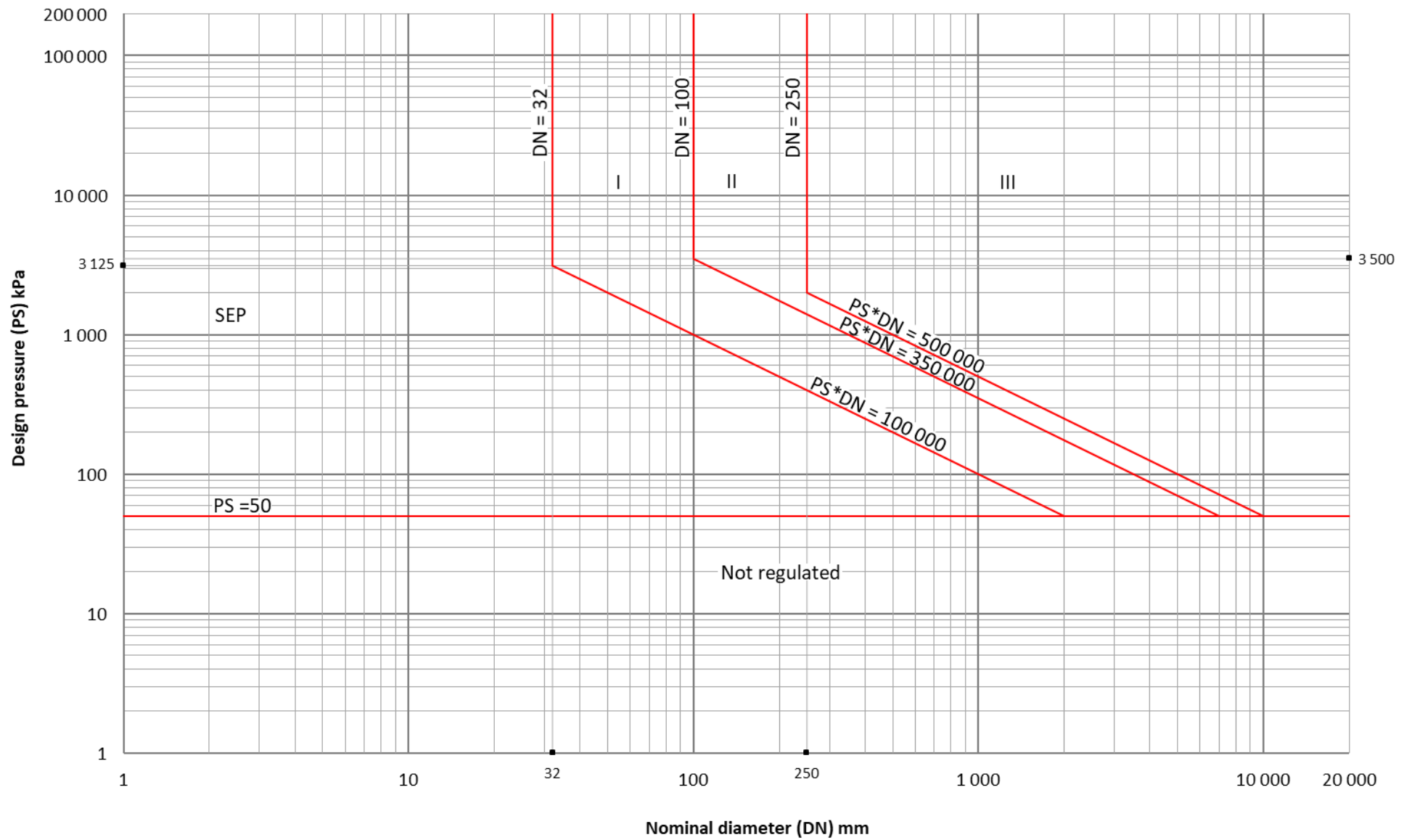


Figure 7 - Graph for piping - Non-dangerous gas

Graph for piping - Dangerous liquids

Figure 8

Defining Equations

PS = 50 000	
X	Y
25	50 000
10 000	50 000

PS* DN = 200 000	
X	Y
25	8 000
2 000	100

PS = 1 000	
X	Y
200	1 000
10 000	1 000

PS = 50	
X	Y
10 000	50
0,1	50

PS = 100	
X	Y
2 000	100
10 000	100

DN = 25	
X	Y
25	8 000
25	200 000

Axis Values

Y-Axis	
X	Y
1	1
1	10
1	100
1	1 000
1	10 000
1	100 000
1	200 000

X-Axis	
X	Y
1	1
10	1
100	1
1 000	1
10 000	1

Additional Minor gridlines

DN = 25(GREY)	
X	Y
25	1
25	8 000

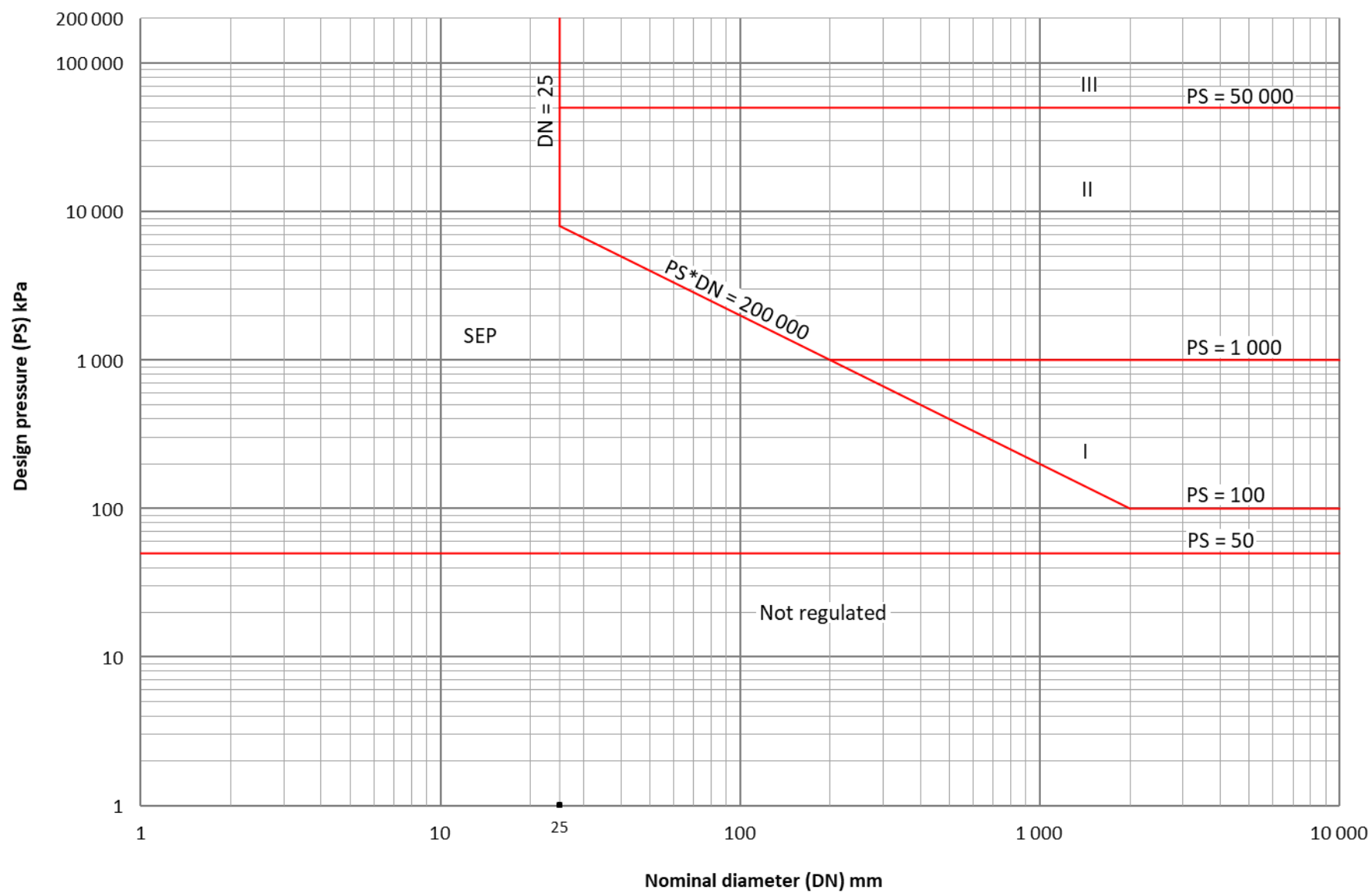


Figure 8 - Graph for piping - Dangerous liquids

Graph for piping - Non-dangerous liquids

Graph 9

Defining Equations

PS = 50		PS*DN = 500 000	
X	Y	X	Y
1	50	200	2 500
10 000	50	500	1 000

PS = 1 000		DN = 200	
X	Y	X	Y
500	1 000	200	2 500
10 000	1 000	200	200 000

PS = 50 000	
X	Y
200	50 000
10 000	50 000

Axis Values

Y- Axis		X- Axis	
X	Y	X	Y
1	1	1	1
1	10	10	1
1	100	100	1
1	1 000	1 000	1
1	10 000	10 000	1
1	100 000		
1	200 000		

Additional Minor gridlines

PS = 2 500	
X	Y
1	2 500
10 000	2 500

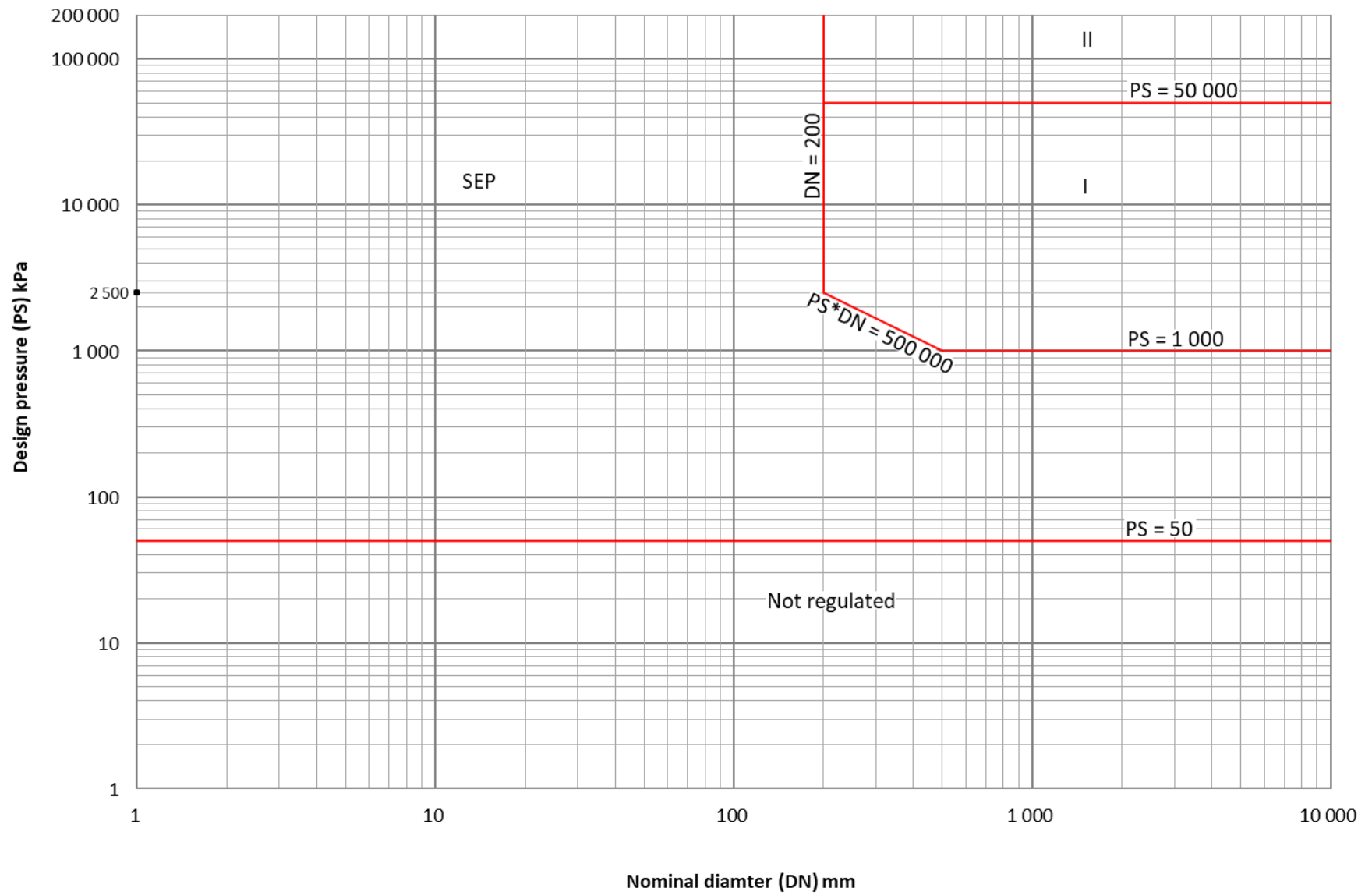


Figure 9 - Graph for piping - Non-dangerous liquids